EE 66 Fall 2026 — Signals, Dynamics, and Information — Discussion 11B

1. Canceling Long Memories

Two filters each remember a little bit of every past input. Can combining their outputs make the resulting system BIBO stable, even if neither filter is stable on its own?

Consider two DT-LTI systems with impulse responses

$$h_1[n] = \begin{cases} \frac{1}{n+1}, & n \geq 0, \\ 0, & n < 0, \end{cases} \qquad h_2[n] = \begin{cases} \frac{1}{n+2}, & n \geq 0, \\ 0, & n < 0. \end{cases}$$

The same input $x[n]$ is passed into both systems, producing outputs $y_1[n]$ and $y_2[n]$.

Recall that a DT-LTI system is *bounded-input, bounded-output (BIBO) stable* if and only if its impulse response is absolutely summable:

$$\sum_{n=-\infty}^{\infty} |h[n]| < \infty.$$

(a) Are the two original systems BIBO stable? Justify your answers using their impulse responses. Both $h_1[n]$ and $h_2[n]$ approach zero as $n \to \infty$; why is this not enough to establish BIBO stability? You may use that the harmonic series $\sum_{k=1}^{\infty} 1/k$ diverges.

We combine these outputs to form a new system:

$$y[n] = a\,y_1[n] + b\,y_2[n], \qquad a, b \in \mathbb{R}.$$

(b) Find the impulse response $h_{a,b}[n]$ of the combined system in terms of a and b. Briefly explain your reasoning.

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(c) Set $a = 1$ and $b = -1$. Simplify the combined impulse response and compute its absolute sum. Is this system BIBO stable? Explain how your answer is consistent with part (a).

Hint: First compute the finite sum from $n = 0$ to $n = N$, then take a limit.

(d) For general a and b, show that the combined impulse response can be written as

$$h_{a,b}[n] = \frac{a+b}{n+2} + \frac{a}{(n+1)(n+2)}, \qquad n \geq 0.$$

Which term has a finite absolute sum for every choice of a and b? Use this expression to find the condition on a and b for the combined system to be BIBO stable.

(e) You intend to subtract the two outputs as in part (c), but a small error in the second coefficient gives $a = 1$ and $b = -1 + \varepsilon$, where $\varepsilon \in \mathbb{R}$. Write the resulting impulse response, then use part (d) to find the values of ε for which the system is BIBO stable.